# Electrochemistry

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ELECTROCHEMISTRY

1. REDOX

OIL RIG (Oxidation Is Loss)

2. OXIDATION NUMBERS
Free element: 0
Monatomic ion: charge
O usually: -2
H usually: +1
Sum in molecule = charge

3. GALVANIC CELL
Anode: oxidation (-)
Cathode: reduction (+)

Cell notation:

EMF: E cell = E cathode - E anode

4. NERNST EQUATION
E = E  - (RT/nF)lnQ
E = E  - (0.0592/n)logQ at 25 C

5. STANDARD REDUCTION POTENTIALS

6. ELECTROLYSIS
Faraday’s laws
F = 96500 C/mol
Used for plating, refining

7. BATTERIES
Dry cell (1.5 V)

Lead-acid (2 V/cell)
Lithium-ion (3.7 V)
Fuel cells